Practice Quiz: Cognition

Part A: Multiple Choice

1. When a crocheter reads a pattern and mentally visualizes the finished fabric, this is best described as: A) A perception error B) Engaging the generative model to generate predictions from symbolic instructions C) External state modification D) The Markov blanket dissolving

2. Mental stitch counting — holding a number in working memory and decrementing with each stitch — is an example of: A) Active inference on the environment B) Internal simulation running the generative model forward in parallel with physical action C) Sensory state processing D) The absence of cognitive activity

3. An experienced crocheter can work simple stitches while holding a conversation because: A) They are not paying attention to the crochet at all B) Their generative model has no predictive power C) Low-level motor models in the hands handle stitch execution automatically, freeing higher-level cognition D) Conversation and crochet use the exact same cognitive resources

4. When a crocheter encounters an unfamiliar stitch and must stop to focus, this shift from automatic to controlled processing occurs because: A) The crocheter has decided to stop crocheting B) Low-level models cannot resolve the prediction error, so it propagates upward to conscious processing C) The Markov blanket has been destroyed D) All cognition is always conscious

5. Counterfactual reasoning in crochet — such as asking "If I add another repeat, will it be wide enough?" — involves: A) Running the generative model in simulation mode to evaluate outcomes of untaken actions B) Randomly guessing the answer C) Only relying on sensory input D) Eliminating the generative model

6. An expert crocheter reads "ch 3, skip 2, dc" and immediately visualizes filet mesh. A beginner reads the same and sees only individual steps. The difference is: A) The expert cannot read pattern notation B) The expert's deeper generative model translates notation

into predicted fabric structure C) The beginner has a more sophisticated model D) Pattern reading requires no generative model

7. "Thinking with your hands" in crochet reflects the Active Inference principle that: A) Cognition occurs only in the brain B) The generative model can be distributed across the body, with local models in the motor system C) Hands have no role in prediction D) Embodied cognition is impossible

Part B: Short Answer

1. Explain how the hierarchical structure of a crochet pattern (project > section > row > stitch) maps onto a hierarchical generative model. Where does high-level cognition operate, and where does low-level automatic processing take over?

2. A crocheter discovers a mistake three rows back that they did not notice at the time. Describe the cognitive process of retroactively identifying what went wrong using the generative model's ability to infer hidden past states.

3. You want to substitute a thicker yarn in a pattern designed for a thinner one. Describe the counterfactual simulation you would need to run to predict how this substitution affects gauge, dimensions, and fabric properties.